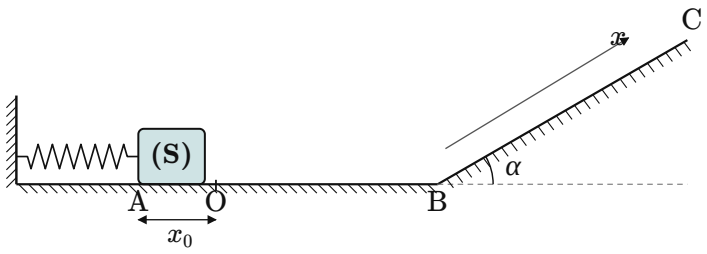
Grade 12 — Life Sciences Lebanese curriculum	**Practice Exam 1** Chapters 1 & 2 — Energy · Linear Momentum	Duration: 2 hours Total: 20 points

This exam is formed of three obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

Exercise 1 — The spring launcher (7 points)

A toy launcher is formed of a spring of stiffness $k = 200\ \text{N/m}$, placed on a horizontal track ABC and fixed from one end. A block (S), considered as a particle of mass $m = 0.5\ \text{kg}$, is placed against its free end. The spring is compressed by $x_0 = AO = 10\ \text{cm}$, O being the position of the free end of the unstretched spring (Doc. 1).



Doc. 1

The horizontal part AB is frictionless; BC is a straight incline making an angle $\alpha = 30°$ with the horizontal. (S) leaves the spring at O.
The horizontal plane through AB is taken as the reference level of gravitational potential energy.

Part 1 — The launch

1.1 (S) is released without initial velocity from A. Calculate the mechanical energy of the system [(S), spring, Earth] at the instant of release.

1.2 Justify that this mechanical energy is conserved between A and O. Deduce that (S) leaves the spring at O with a speed $V_0 = 2\ \text{m/s}$.

Part 2 — Climbing the incline without friction

In this part, friction on BC is neglected. A point M of BC is located by its abscissa $x = BM$ along the incline; the speed of (S) at M is v. (S) reaches B with the speed V_0.

2.1 Write the expression of the mechanical energy of the system [(S), Earth] at M in terms of m, v, g, x and α.

2.2 Deduce the maximum distance L covered by (S) on BC.

2.3 By deriving the mechanical energy with respect to time, show that the acceleration of (S) is $a = -g\sin\alpha$. Deduce the nature of the motion of (S) on BC.

Part 3 — The real incline

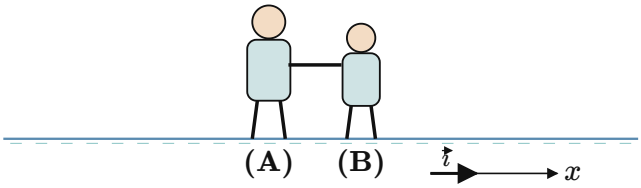
In reality, (S) is acted upon on BC by a constant friction force $\vec{f}$ parallel to the incline. (S) stops at a point D such that $BD = 25$ cm.

3.1 Calculate the variation ΔME of the mechanical energy of the system [(S), Earth] between B and D.

3.2 Deduce the magnitude f of the friction force and the variation ΔU of the internal energy of the system [(S), incline, Earth].

Exercise 2 — Two skaters push each other (7 points)

Two ice-skaters, (A) of mass $m_A = 60$ kg and (B) of mass $m_B = 40$ kg, are standing at rest face to face on a horizontal ice rink (Doc. 2). They push each other with their hands. Friction on the ice is neglected.



Doc. 2

Part 1 — The system [(A), (B)]

1.1 Name the external forces acting on the system [(A), (B)]. Show that this system is pseudo-isolated.

1.2 Give the value of the linear momentum $\vec{P}$ of the system before the push. Deduce its value $\vec{P}'$ just after the push.

Part 2 — Just after the push

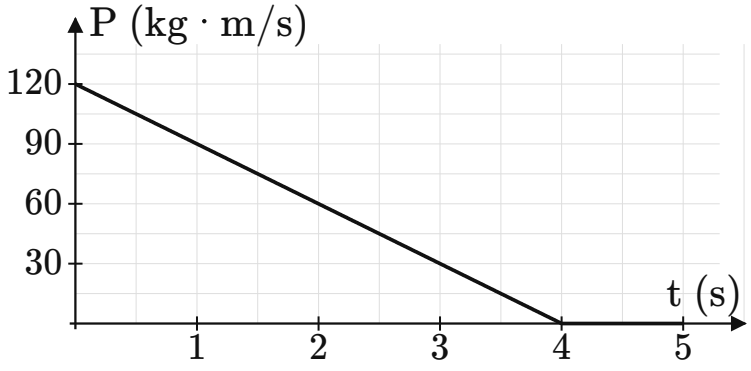
Just after the push, (A) moves with the velocity $\vec{v}_A = -2\,\vec{i}$ (m/s).

2.1 Determine the velocity $\vec{v}_B$ of (B) just after the push.

2.2 Calculate the kinetic energy of the system [(A), (B)] just after the push. Specify the origin of this energy.

Part 3 — (B) on a rough zone

(B) then enters a rough zone of the rink at the instant $t_0 = 0$, where it is acted upon by a constant horizontal friction force $\vec{f}$. Doc. 3 shows the magnitude P of the linear momentum of (B) as a function of time.



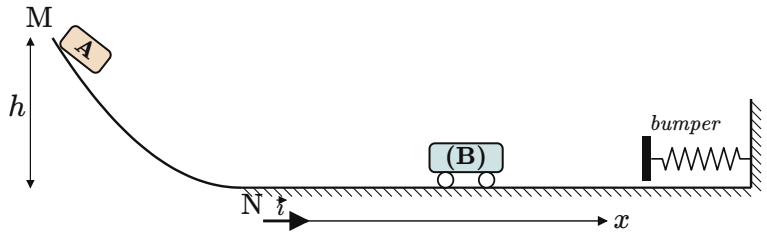
Doc. 3

3.1 Using Doc. 3 and Newton's second law $\Sigma\vec{F} = \dfrac{d\vec{P}}{dt}$, determine the magnitude f of the friction force.

3.2 By applying the variation of the mechanical energy of the system [(B), Earth], determine the distance d covered by (B) on the rough zone until it stops.

Exercise 3 — Two carts and a bumper (6 points)

A cart (A) of mass $m_A = 1$ kg is released without initial velocity from a point M of a ramp, at a height $h = 45$ cm above the horizontal track (Doc. 4). A cart (B) of mass $m_B = 2$ kg is at rest on the track. A bumper, a spring of stiffness $k = 300$ N/m, is fixed at the end of the track. Friction is neglected everywhere; the carts are considered as particles.



Doc. 4

> The horizontal plane of the track is the reference level of gravitational potential energy.

Part 1 — Down the ramp

1.1 Justify that the mechanical energy of the system [(A), Earth] is conserved. Calculate the speed v_A of (A) at the bottom N of the ramp.

Part 2 — The collision

(A) hits (B); the two carts stick together and move with a common velocity $\vec{V} = V\,\vec{i}$.

2.1 The system [(A), (B)] is pseudo-isolated during the collision. Determine V.

2.2 Calculate the kinetic energy of the system [(A), (B)] before and after the collision. Deduce the type of this collision.

Part 3 — The bumper

3.1 The two carts compress the bumper. Calculate the maximum compression X of the spring.

3.2 The carts leave the bumper with the velocity $\vec{V}' = -V\,\vec{i}$. Calculate the variation $\Delta\vec{P}$ of the linear momentum of the carts. Deduce the average force exerted by the bumper on the carts, knowing that the contact lasts $\Delta t = 0.2$ s.

— End of the exam —